DAY 1

From Raw Data to Clean Cells

QC · Filtering · Setting the stage for biology

1 Load Raw Data

Read the 10x H5 file into a Seurat object. Inspect dimensions: genes × barcodes.

`Read10X_h5() → CreateSeuratObject()` ~33,500 genes × 700k barcodes

2 Compute QC Metrics

Calculate genes per cell, UMIs per cell, and mitochondrial read percentage.

`PercentageFeatureSet(pattern="^MT-")`

nFeature_RNA
200–3,000 → real cell

percent.mt >20%
→ dying / stressed

3 Visualise QC Distributions

Violin plots and scatter plots. Decide thresholds before filtering — data first, numbers second.

`VlnPlot() + FeatureScatter()` nCount vs nFeature scatter · percent.mt violin

4 Filter Low-Quality Cells

Apply thresholds to remove empty droplets, doublets, and dead cells.

`subset(nFeature_RNA > 200 & < 3000 & percent.mt < 10)`

✓ 1,700 high-quality cells retained

⚠ ~98.8% barcodes removed (mostly empty)

Overnight question
What does a cluster represent biologically? How might two cells end up in the same cluster?

✅ **Day 1 output:** A clean Seurat object (1,700 cells) with QC metrics computed, thresholds justified, and ready for normalisation.

DAY 2

From Clean Cells to Cell Identity

Normalise · Cluster · Annotate · Answer

1 Normalise

Scale each cell to 10,000 UMIs, then log-transform so depth differences don't confound biology.

`NormalizeData()` Comparable expression values across all cells

2 Find Variable Genes

Select top 2,000 genes that vary most cell-to-cell. Housekeeping genes are excluded — they carry no cell-type signal.

`FindVariableFeatures(nfeatures=2000)` 2,000 informative genes (from 33,500)

3 Scale → PCA

Center and scale genes, then compress 2,000 dimensions into top 30 principal components capturing biological variance.

`ScaleData() → RunPCA(npcs=50)`

PC1 ≈ lymphoid vs myeloid

PC2 ≈ T vs B cells

4 UMAP Visualisation

Project cells into 2D for visual inspection. Nearby cells = transcriptionally similar. Global distances are not meaningful.

`RunUMAP(dims=1:30)` 2D map of cell relationships

5 Cluster Cells

Build a nearest-neighbour graph, then detect communities. Resolution 0.5 gives 12 clusters for this dataset.

`FindNeighbors() → FindClusters(res=0.5)` 12 clusters · Cluster 0 = 279 cells (largest)

6 Find Marker Genes

For each cluster, find genes that are highly and specifically expressed. Sort by avg_log2FC with low pct.2.

`FindAllMarkers(only.pos=TRUE)`

Cluster 0 top: LEF1, FHIT, MAL

Cross-ref canonical markers

Annotate & Answer

LEF1, FHIT, MAL → canonical Naive CD4+ T cell markers. Cluster 0 (279 cells) is the most abundant.

Answer: Naive CD4+ T cells

🏆 **Workshop answer:** Cluster 0 = **Naive CD4+ T cells** — largest cluster, driven by LEF1/FHIT/MAL, consistent with expected PBMC composition.

Note on the figures: These diagrams are designed to be printed or embedded in slides. Each step maps directly to a slide in the workshop deck. The code snippets use simplified R syntax for readability — full code is in `workshop.R`.